The Big Bang



Use the information below to answer your True/False questions:

How many did you get right? _____

Using a piece of A3 paper, make five large diagrams depicting each of the paragraphs below: Use pictures and notes to describe what is happening:

Most astronomers believe the Universe began in a Big Bang about 14 billion years ago. At that time, the entire Universe was inside a bubble that was thousands of times smaller than a pinhead. It was hotter and denser than anything we can imagine.

Ideas: E.g. Heading – 14 billion years ago! Large picture of a pin. Arrow pointing to the head. Labelled 'whole universe smaller than this!' Flames coming off the pinhead. Hand trying to hold up pin – showing it is very heavy!

Then it suddenly exploded. The Universe that we know was born. Time, space and matter all began with the Big Bang. In a fraction of a second, the Universe grew from smaller than a single atom to bigger than a galaxy. And it kept on growing at a fantastic rate. It is still expanding today.

Ideas:_____
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As the Universe expanded and cooled, energy changed into particles of matter and antimatter. These two opposite types of particles largely destroyed each other. But some matter survived. More stable particles called protons and neutrons started to form when the Universe was one second old.

Ideas:_____
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Over the next three minutes, the temperature dropped below 1 billion degrees Celsius. It was now cool enough for the protons and neutrons to come together, forming hydrogen and helium nuclei.

Ideas:_____
_____
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_____

After 300 000 years, the Universe had cooled to about 3000 degrees. Atomic nuclei could finally capture electrons to form atoms. The Universe filled with clouds of hydrogen and helium gas.

Ideas:_____
_____
_____
_____





The Birth of Galaxies

Use the information below to answer your True/False questions:

How many did you get right? _____

Using a piece of A3 paper, make five large diagrams depicting each of the paragraphs below: Use pictures and notes to describe what is happening:

We cannot see anything that happened during the first 300 000 years of the Universe. Scientists try to work it out from their knowledge of atomic particles and from computer models.

Ideas:_____
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The only direct evidence of the Big Bang itself is a faint glow in space. Spacecraft and telescopes on balloons see this as a patchy pattern of slightly warmer and cooler gas all around us. These ripples also show where the hydrogen clouds were slightly denser.

Ideas:_____
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As millions of years passed, the dense areas pulled in material because they had more gravity. Finally, about 100 million years after the Big Bang, the gas became hot and dense enough for the first stars to form.

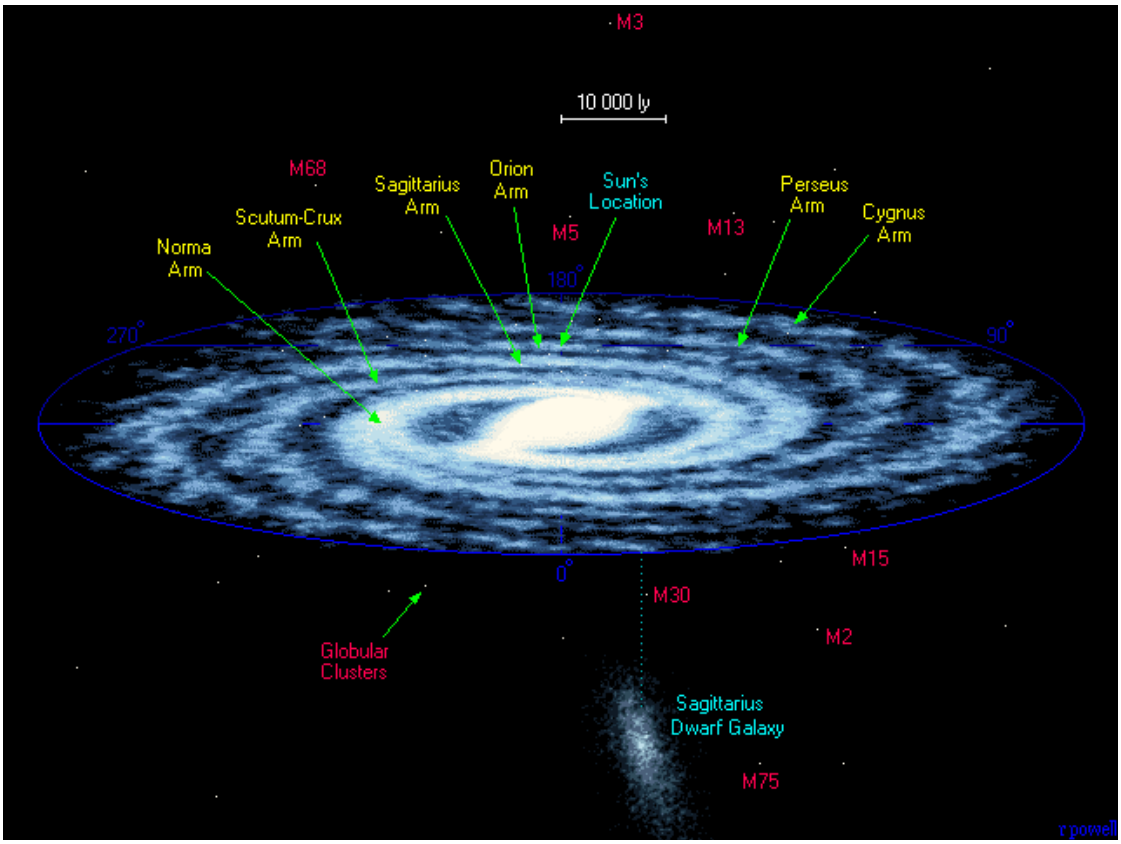
Ideas:_____
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New stars were being born at a rate 10 times higher than in the present-day Universe. Large clusters of stars soon became the first galaxies.

Ideas:_____
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_____

The Hubble Space Telescope and powerful ground-based telescopes are now beginning to find galaxies that were created about one billion years after the Big Bang. These small galaxies were much closer together than galaxies are today. Collisions were common. Like two flames moving towards each other, they merged into bigger galaxies. Our Milky Way galaxy came together in this way.

Ideas:_____
_____
_____
_____



In the beginning



Use the information below to make **5** true or false questions:

Swap with someone in the class and answer their questions:

How many did you get right without looking back at the notes! _____

Using a piece of A3 paper, make five large diagrams depicting each of the paragraphs below: Use pictures and notes to describe what is happening:

Living things breathe, feed, grow and leave offspring. They range from the simplest single-celled bacteria to plants, birds, animals and humans. Earth is the only world we know that supports life. No one knows how life began. It seems to involve the coming together of certain chemicals. But what changes these chemicals into life-forms is a mystery.

Ideas:_____
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_____

Studies of ancient rocks have shown that life began on Earth about 3 800 million years ago. At that time, the Earth was a very unfriendly place. The air was dense and poisonous. Comets and meteorites were raining down on the planet.

Ideas:_____
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The young Earth was a water world with small areas of dry land. Many scientists think that life began in the lakes and oceans. Many of the ingredients from which life started may have been brought by comets and meteorites.

Ideas:_____
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Did life begin in the oceans?

Did life begin in the oceans?

Lightning and ultraviolet light from the Sun may have split the hydrogen-rich gases in the atmosphere. Energy may also have come from hot springs on the sea bed, known as ‘black smokers’. Over time, the simple chemicals joined to become larger, more complex chemicals.

Ideas:_____
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Once these chemicals collected in the oceans or lakes, they formed an 'organic soup'. One day, an accident occurred. A molecule began to make copies of itself. Earth had sprung to life.

Ideas:_____
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_____
_____



The Big Bang

Answer the following questions **True** or **False**:

1. The universe began 14 000 million years ago. _____

2. The universe came from something thousands of times smaller than a pinhead smaller. _____

3. It took years for the universe to grow from smaller than a single atom to bigger than a galaxy. _____

4. The universe has stopped growing. _____

5. As the universe cooled energy changed into ‘matter’ and ‘you don’t matter’.
_____

6. Protons and neutrons formed when the universe was 1 second old.
_____

7. The temperature dropped to below 1 billion degrees Celsius in 3 seconds.
_____

8. Hydrogen and Helium were the first elements formed.
_____

9. After 3000 years the universe had cooled to 3000°C.
_____



The Birth of Galaxies

Answer the following questions **True** or **False**:

1. We have photos of what happened during the first 300 000 years of the universe. _____

2. Scientists work out the first 300 000 years from watching super models.

_____

3. A faint glow in space is the only evidence of the Big Bang Theory.

_____

4. Slightly warmer and cooler gas can be seen from spacecraft and telescopes.

_____

5. Over millions of years material was pulled into denser areas by gravity.

_____

6. In the beginning stars were being born at a rate 100 times faster than present.

_____

7. Clusters of stars became planets.

_____

8. Powerful telescopes can find galaxies that were created over a billion years ago.

_____

9. Our galaxy is the Snickers galaxy.

_____

